



Republic of the Philippines

Department of Environment and Natural Resources

MINES AND GEOSCIENCES BUREAU

North Avenue, Diliman, Quezon City, Philippines

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FLOOD THREAT ADVISORY

To: **Brgy. Chairperson**
Brgy. Concepcion Uno
City of Marikina
Metropolitan Manila

Date: **October 9, 2024**

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Spain-Fairlane-JP Rizal	14° 39' 51.1" N, 121° 06' 21.3" E 14° 39' 51.8" N, 121° 06' 16.8" E	Moderate to High Moderate to High	In the northern portion of the barangay, Spain St. and its vicinity experienced flood heights greater than 1m during Ondoy, and around 1m during Ulysses. After a few hours, the flooding has subsided. Lower flood heights were noted in the area of Fairlane cor. JP Rizal, with less than 0.5m reported during both events. A previously active channel of Marikina River was reported to be filled with material to be rerouted, which is also complemented by a recent improvement of the drainage system in the corner of JP Rizal and Fairlane to further curb the effects of flooding in the area.
Goodrich	14° 39' 40.1" N, 121° 06' 15.0" E	High	The Balante Creek is proximal to the observation point in Goodrich and is reportedly inundating the surrounding areas with maximum of ~1.4m-high floodwaters during extreme rainfall events (Ondoy) and roughly 1m during 2012 Habagat season and Ulysses.
Libis	14° 39' 16.5" N, 121° 06' 04.5" E	Very High	Libis is relatively depressed compared to the main JP Rizal St. Due to this and its proximity to Ampalaya Creek, maximum flood heights during Ondoy were reported to be greater than 2m, which also reached the elevated highway. The difference between that event and Ulysses, is that floodwaters did not reach the highway during the latter.
San Isidro I (P. Gomez St.)	14° 39' 10.6" N, 121° 06' 22.6" E	High	The confluence of Balante and Antipolo Creeks is found in San Isidro I / P. Gomez St., which inundated surrounding areas with ~1m-high floodwaters during Ondoy and 0.5m during Ulysses. Floodwaters immediately subside after the rain had stopped.
San Isidro II (Balante)	14° 39' 07.3" N, 121° 06' 22.8" E	High	Balante Creek inundates the surrounding area of San Isidro II with varying maximum flood heights, ranging

Creek)	14° 39' 05.1" N, 121° 06' 23.2" E	Moderate	from 0.5 to 1.5m during Ondoy. Floodwaters subside immediately after the rain had stopped. Also noted during the assessment is a repaired section of the riprap, whose rubble remain on the riverbed.
	14° 39' 01.4" N, 121° 06' 22.1" E	High	
Antipolo Creek	14° 39' 13.4" N, 121° 06' 31.6" E	High	Traversing the upstream section of Antipolo Creek (prior to joining Balante Creek), patches of sandbars were noted along the river bed, as well as a previously damaged, now repaired, section of riprap. Cracks were also observed along the pathway. The proximal area of Simeona Village (Givenchy) also reported roughly 1m-high floodwaters during Ondoy, and none during Ulysses.

Remarks / Recommendations:

- ✓ It is highly recommended to implement appropriate easements in construction of settlements or other infrastructure along rivers/creeks as specified in the Presidential Decree No. 1067 known as Water Code of the Philippines.
- ✓ Improvement and proper maintenance of existing drainage system in the barangay are recommended to prevent inundation of residential areas, critical facilities, and other assets. Avoid improper waste disposal, particularly along drainage canals as it may cause clogging and obstruction of water flow. Ensure regular clearing of natural and artificial drainage pathways, or prior to an impending tropical cyclone or other rain-bearing weather systems.
- ✓ Construction of appropriate and well-studied engineering measures (e.g., dike, flood wall, etc.) should be implemented to minimize the rate of erosion along rivers/creeks. Ensure that the structure follows engineering standards and has structural integrity. Also, damages in the structural mitigation measures should be reported to concerned agency/office responsible for the installation. The LGU is also encouraged to maintain such measures or reinforce identified damages, with the strong support of the barangay officials and active engagement of the communities and the private sector.
- ✓ Install a warning signage alerting motorists and residents of the danger of crossing bridges susceptible to flash flood during heavy rainfall events.
- ✓ The CDRRMO, together with the BDRRMO, should conduct Information, Education, and Communication (IEC) Campaign on flood hazards to flood-prone communities, particularly the households residing proximal to the river and irrigation canals.
- ✓ Strengthen and improve flood warning system with corresponding warning levels for pre-emptive evacuation purposes. The barangay should have constant communication with nearby and/or upland/upstream barangays during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas.
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of tropical cyclones and other prevailing weather disturbances. This may include City Disaster Risk Reduction Management Office (CDRRMO), Office of the Civil Defense (OCD) NCR, and other related agencies.

Prepared by:



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Received by:
Designation:
Office:

BARANGAY CONCEPCION UNO



In the northern portion of the barangay, Spain St. (*left*) and its vicinity experienced flood heights greater than 1m during Ondoy, and around 1m during Ulysses. After a few hours, the flooding has subsided (GPS Coordinates: 14° 39' 51.1" N, 121° 06' 21.3" E). Lower flood heights were noted in the area of Fairlane cor. JP Rizal (*right*), with less than 0.5m reported during both events. A previously active channel of Marikina River was reported to be filled with material to be rerouted, which is also complemented by a recent improvement of the drainage system in the corner of JP Rizal and Fairlane to further curb the effects of flooding in the area (GPS Coordinates: 14° 39' 51.8" N, 121° 06' 16.8" E).



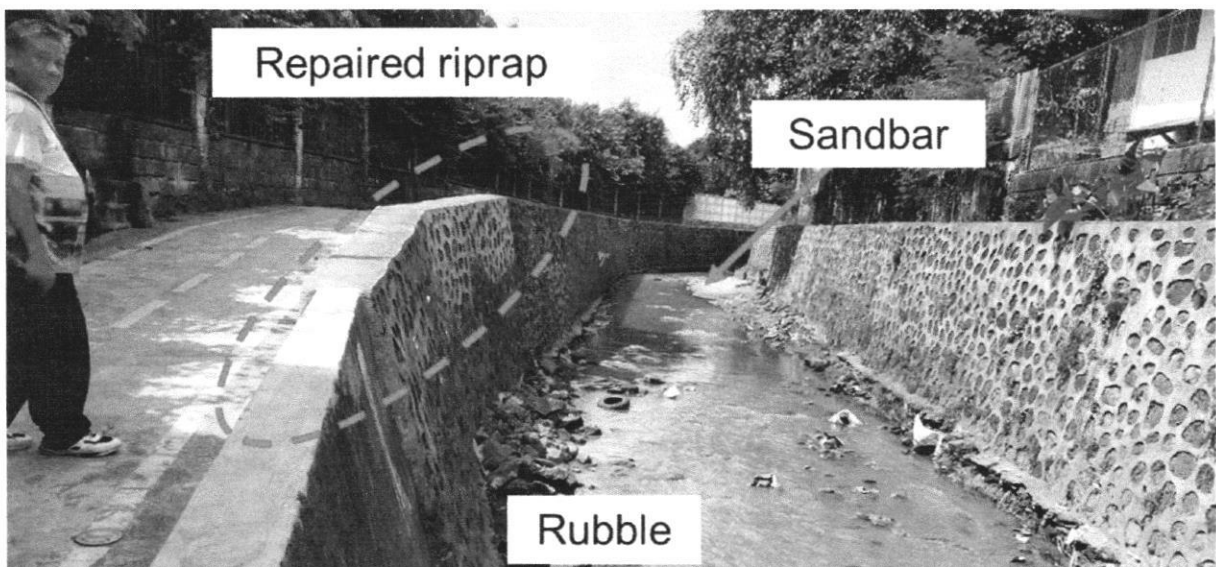
The Balante Creek is proximal to the observation point in Goodrich (*left*) and is reportedly inundating the surrounding areas with maximum of ~1.4m-high floodwaters during extreme rainfall events (Ondoy) and roughly 1m during 2012 Habagat season and Ulysses (GPS Coordinates: 14° 39' 40.1" N, 121° 06' 15.0" E). While the area on the *right* (Libis) is relatively depressed compared to the main JP Rizal St. Due to this and its proximity to Ampalaya Creek, maximum flood heights during Ondoy were reported to be greater than 2m, which also reached the elevated highway. The difference between that event and Ulysses, is that floodwaters did not reach the highway during the latter (GPS Coordinates: 14° 39' 16.5" N, 121° 06' 04.5" E).



The confluence of Balante and Antipolo Creeks is found in San Isidro I / P. Gomez St., which inundated surrounding areas with ~1m-high floodwaters during Ondoy and 0.5m during Ulysses. Floodwaters immediately subside after the rain had stopped (GPS Coordinates: 14° 39' 10.6" N, 121° 06' 22.6" E).



Balante Creek inundates the surrounding area of San Isidro II (*top*) with varying maximum flood heights, ranging from 0.5 to 1.5m during Ondoy. Floodwaters subside immediately after the rain had stopped (GPS Coordinates: 14° 39' 07.3 and 05.1" N, 121° 06' 22.8 and 23.2" E). Also noted during the assessment is a repaired section of the riprap (*bottom*), whose rubble remain on the riverbed (GPS Coordinates: 14° 39' 01.4" N, 121° 06' 22.1" E).



Traversing the upstream section of Antipolo Creek (prior to joining Balante Creek), patches of sandbars were noted along the river bed, as well as a previously damaged, now repaired, section of riprap. Cracks were also observed along the pathway. The proximal area of Simeona Village (Givenchy) also reported roughly 1m-high floodwaters during Ondoy, and none during Ulysses (GPS Coordinates: 14° 39' 13.4" N, 121° 06' 31.6" E).

Recommendations:

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